

Electromechanical Characterization of a Piezoelectric Ring

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Abstract

This report details the experimental characterization of a piezoelectric ring structure instrumented with 15 piezoelectric patches. The experiment focused on 3 batches (A, B, C), each intended to group 4 patches. The capacitance for a full 4-patch batch was measured at 135 nF (Batch A) and 133 nF (Batch C). A lower value for Batch B (103 nF) was later attributed to a faulty connection (resulting in 3 active patches). The Frequency Response Functions (FRFs) of the ring were measured for the first three vibration modes under two main test sets (short-all/open-one, and open-all/short-one) using the DEWESoftX data acquisition system. Subsequent analysis in MATLAB, based on the resonant frequency under open-circuit (f_O) and short-circuit (f_S) conditions, yielded k_c , L , and R . Experimental difficulties, primarily the undetected faulty patch, resulted in data requiring strict filtering, with only 7 out of 18 total data pairs meeting the fundamental condition ($f_S < f_O$). The calculated coupling factors were consistently low, ranging from 0.88% to 3.31%, confirming $k_c \leq 5\%$, indicating limited electromechanical conversion efficiency in the measured configuration.

1 Introduction

Piezoelectric materials, which exhibit the electromechanical coupling effect (converting mechanical strain into electrical charge and vice-versa), are foundational components in the field of smart structures. When bonded to a mechanical structure, these materials can be used for vibration sensing, energy harvesting, or active damping, often utilizing a technique known as piezoelectric shunt damping.

The dynamic behavior of a piezoelectric patch and its host structure around a mechanical resonance can be accurately modeled by a simple electrical R-L-C circuit connected in series with the inherent electrical capacitance of the patch (C^e). The parameters of this circuit the equivalent inductance (L), resistance (R), and the mechanical damping are directly related to the structure's modal properties.

The primary objective of this experiment was to measure the first three mechanical vibration modes of the piezoelectric ring and determine these equivalent electrical parameters (k_c, L, R) for various patching configurations. This characterization is crucial for the subsequent design of effective passive or active piezoelectric shunt circuits.

As per the lab instructor's directive to ensure a comprehensive system analysis, this report evaluates all identifiable vibrational modes rather than focusing on a single assigned mode.

2 Methodology

2.1 Experimental Setup and Data Acquisition

The experimental object was a piezoelectric ring instrumented with a total of 15 piezoelectric patches. For this experiment, 3 batches (A, B, and C) were used, each intended to group 4 patches. The

inherent capacitance (C^ϵ) for each batch was measured. Batch A (4 patches) was 135 nF and Batch C (4 patches) was 133 nF. Batch B measured only 103 nF, a discrepancy later discovered to be from an unconnected patch (meaning it only had 3 active patches instead of 4).

1. **Reference Measurement:** Before data acquisition, a functional check was performed by shorting all piezoelectric circuits via a breadboard (Figure 2c) to establish a baseline. Care was taken to avoid overloading the system.
2. **Instrumentation:** The structural dynamics were excited using an impact hammer (Figure 2b). The ring was struck on its outer surface at a central point diametrically opposite to the accelerometer's placement. The resulting acceleration response was measured using an accelerometer placed on the inner surface of the ring. The overall setup is shown in Figure 1. The force and acceleration signals were recorded using the DEWESoftX data acquisition system (Figure 2a) to compute the Frequency Response Function (FRF). Key settings for the FRF measurement included a resolution of 30000 spectral lines (resulting in a frequency resolution df of 0.167 Hz) and Linear averaging.
3. **Electrical Boundary Conditions:** FRFs were acquired in a systematic two-stage process for the first three vibration modes:
 - **Short-All Tests:** First, all batches were short-circuited. The FRFs were then measured for the "Open A," "Open B," and "Open C" configurations one after another.
 - **Open-All Tests:** Second, all batches were left in open-circuit. The FRFs were then measured for the "Short A," "Short B," and "Short C" configurations one after another.

This process resulted in 6 distinct test configurations, with modes 1, 2, and 3 studied for each, yielding a total of 18 data pairs.

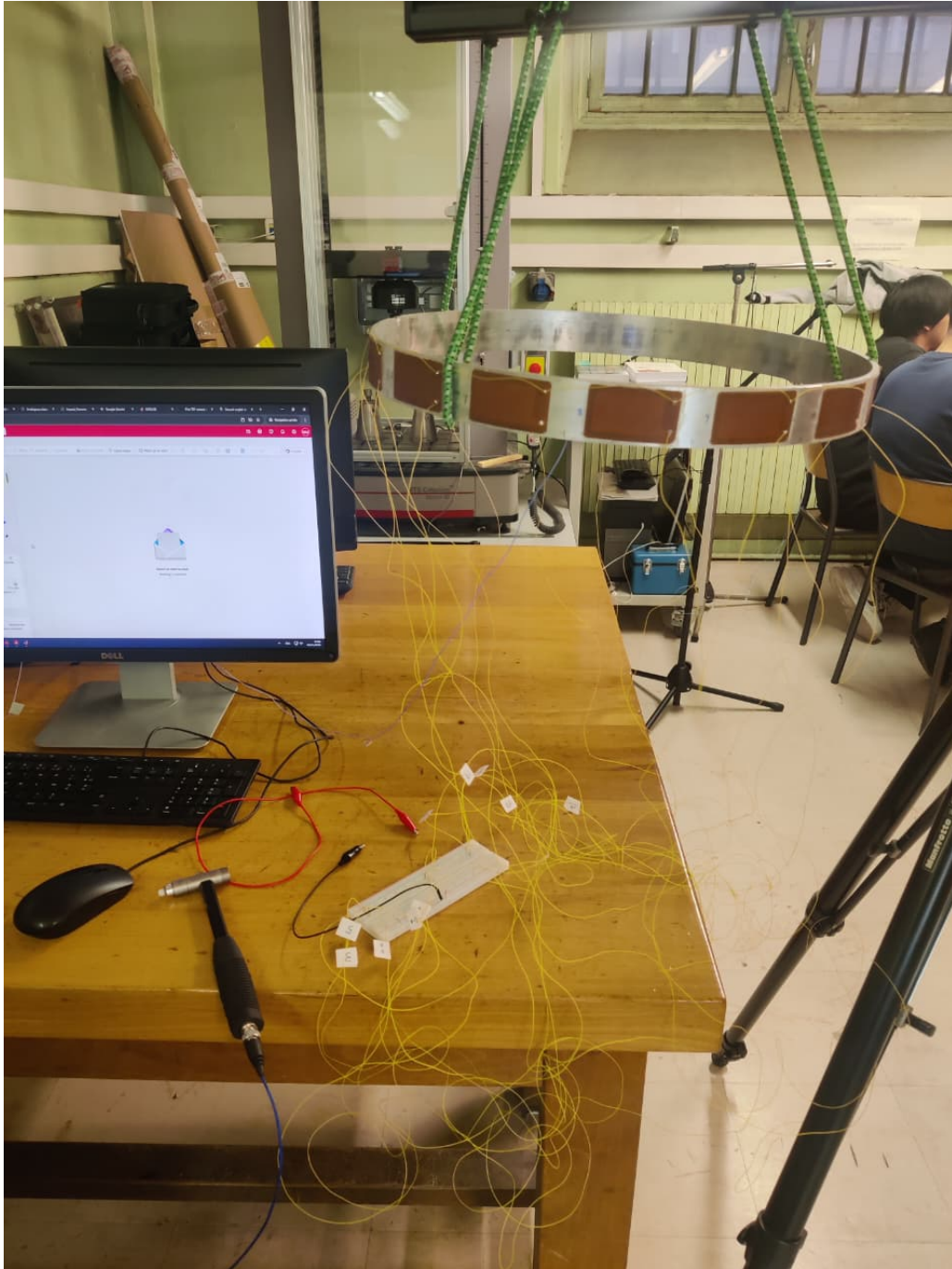


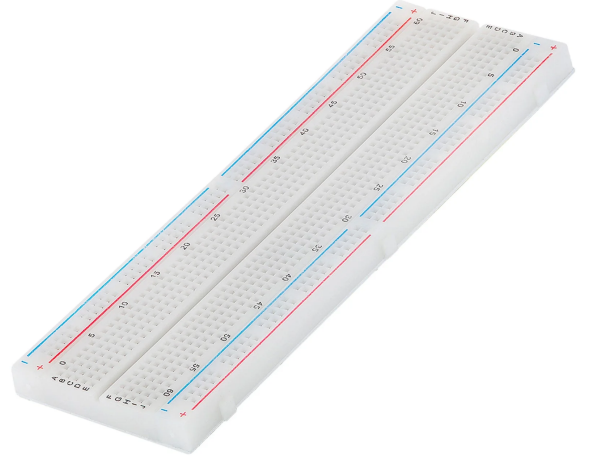
Figure 1: The experimental setup, showing the suspended piezoelectric ring, impact hammer (bottom left), and wiring to the breadboard. The accelerometer was placed on the ring (hidden from view).



(a) DEWE-43A data acquisition system



(b) Impact hammer



(c) MB-102 breadboard

Figure 2: Key instrumentation and components used in the experiment.

2.2 Capacitance Measurement Procedure

The baseline capacitance (C^ϵ) for each piezoelectric batch, mentioned in the previous section, was determined using a high-precision LCR meter. The procedure involved:

- **Connection:** The meter's positive (red) probe was connected to the PZT patch electrode, and the negative (black) probe was connected to the ground.
- **Frequency:** The measurement frequency was set to 1 kHz. This value was chosen as it falls

approximately midway between the primary modes of interest, avoiding the resonant regions where capacitance measurements can be unstable.

2.3 Data Analysis

The recorded FRFs were imported into MATLAB for post-processing. The equivalent circuit parameters were determined using the relationship between the open-circuit resonant frequency (ω_O) and the short-circuit resonant frequency (ω_S).

The electromechanical coupling factor (k_c) was calculated using the formula:

$$k_c = \sqrt{\frac{\omega_O^2 - \omega_S^2}{\omega_S^2}}$$

The equivalent inductance (L) was calculated from the open-circuit frequency and the inherent capacitance C^ϵ :

$$L = \frac{1}{C^\epsilon \omega_O^2}$$

The equivalent resistance (R) was calculated using the formula:

$$R = \sqrt{\frac{3}{2}} \frac{k_c}{C^\epsilon \omega_O}$$

Note: Where $\omega_O = 2\pi f_O$ and $\omega_S = 2\pi f_S$ are the angular frequencies.

3 Modal Damping Ratios (ζ)

Damping ratios were identified in MATLAB using `modalfit` (peak-picking with least-squares rational function refinement). Table 1 reports ζ for each configuration and mode under open- and short-circuit conditions. Values are shown as percent. Note that ζ is reported even when a pair later fails the validity check $f_S < f_O$ (and thus is excluded from the k_c, L, R computation).

Table 1: Identified damping ratios ζ from `modalfit` (percent).

Pair	Configuration	Mode	ζ_O (%)	ζ_S (%)
1	Open A / Short All	1	0.61	0.58
		2	0.01	7.11
		3	0.01	7.11
2	Open B / Short All	1	1.14	0.58
		2	0.02	7.11
		3	0.02	7.11
3	Open C / Short All	1	0.59	0.58
		2	0.09	7.11
		3	0.09	7.11
4	Open All / Short A	1	0.61	0.67
		2	0.12	0.35
		3	0.12	0.35
5	Open All / Short B	1	0.61	0.27
		2	0.12	0.02
		3	0.12	0.02
6	Open All / Short C	1	0.61	0.54
		2	0.12	0.54
		3	0.12	0.54

4 Frequency Response Plots (Open vs Short)

The following figures show the magnitude of the FRFs (Open vs Short) exported from MATLAB as vector PDFs.

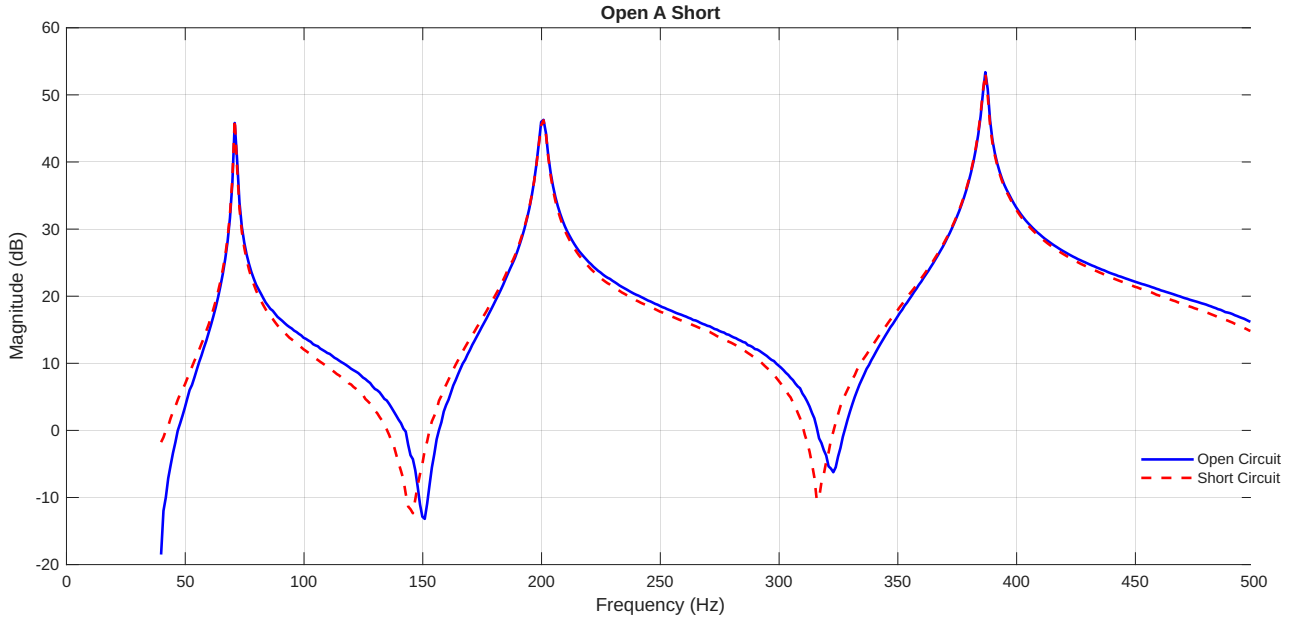


Figure 3: FRF comparison: Open A vs Short.

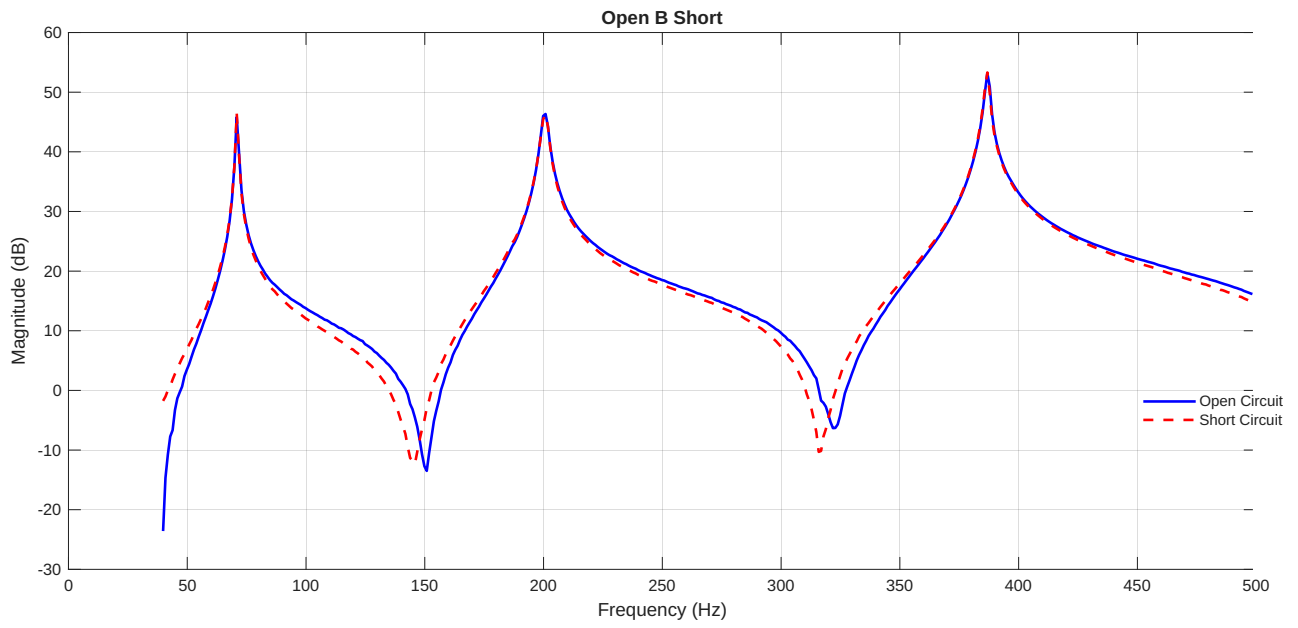


Figure 4: FRF comparison: Open B vs Short.

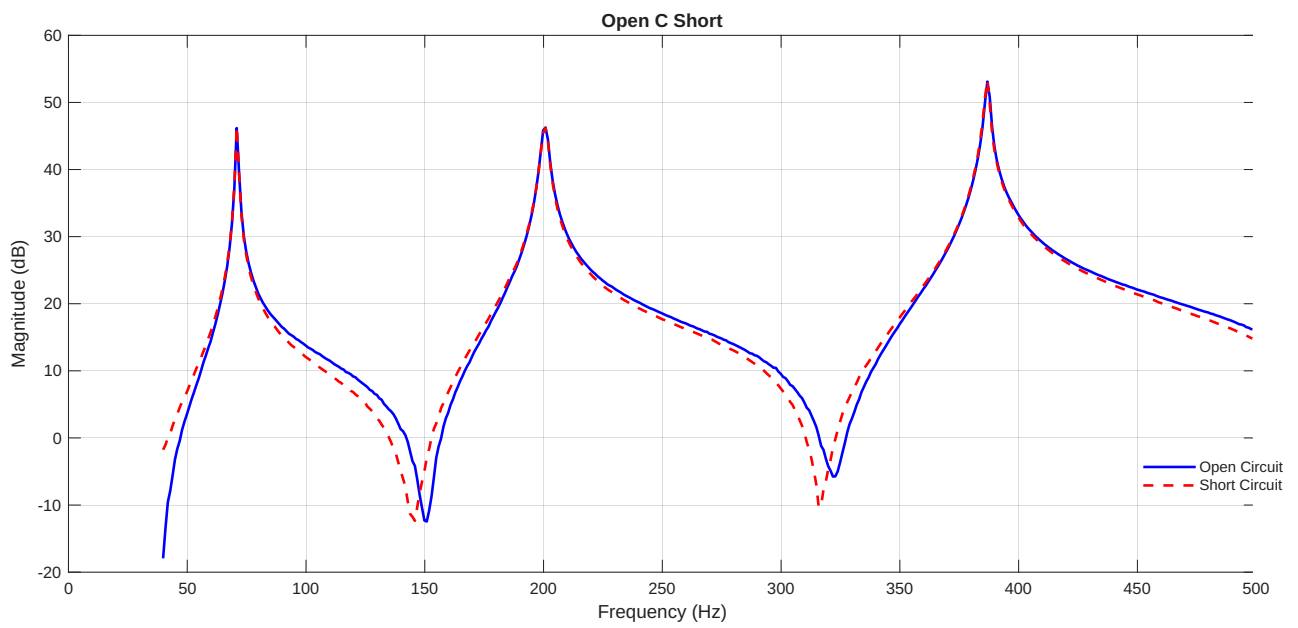


Figure 5: FRF comparison: Open C vs Short.

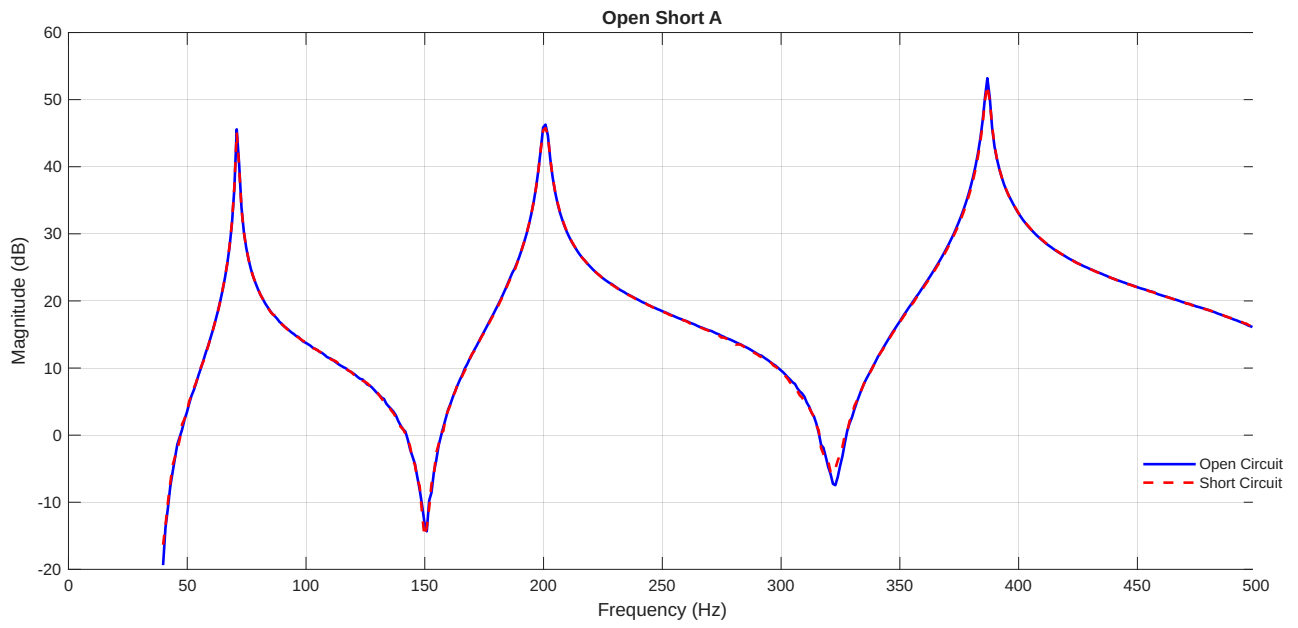


Figure 6: FRF comparison: Open vs Short A.

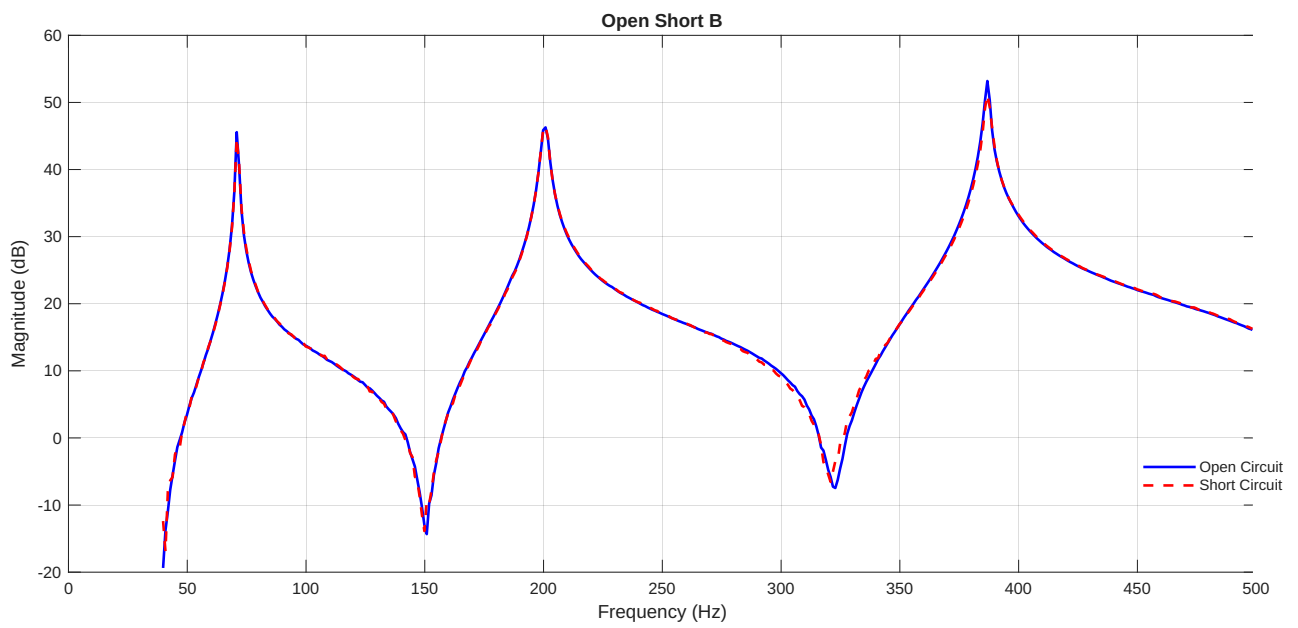


Figure 7: FRF comparison: Open vs Short B.

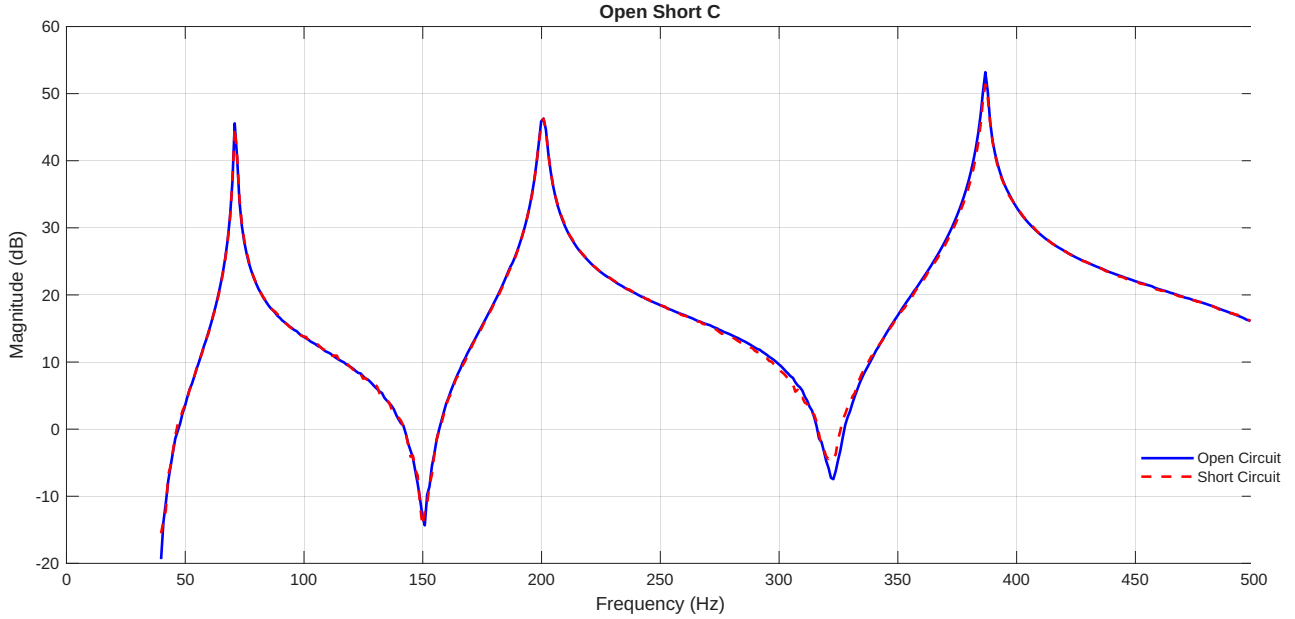


Figure 8: FRF comparison: Open vs Short C.

5 Comments

5.1 Experimental Difficulties

Two significant issues were encountered during the experiment:

1. **Faulty Connection:** A crucial difficulty was the discovery, at the end of the data acquisition, that one of the piezoelectric patches in Batch B had a detached or disconnected wire connection. This meant Batch B was operating with only 3 active patches instead of the intended 4. As noted in the methodology, this fault was responsible for Batch B's low 103 nF capacitance (compared to 135 nF and 133 nF for Batches A and C) and introduced spurious electrical behavior into some configurations.
2. **Data Invalidation:** The core principle for determining k_c in a passive mechanical system is that the short-circuit resonance frequency must be lower than the open-circuit anti-resonance frequency, i.e., $f_S < f_O$. Out of the **18 data pairs measured (6 configurations $\times$ 3 modes)**, this condition was not met for 11 pairs (including all pairs for Mode 2), indicating that the experimental setup was not behaving as an ideal piezoelectric shunt system. Consequently, only the 7 pairs listed in Table 2 were deemed valid for k_c calculation.

5.2 Results Summary

Analysis was attempted on all 18 data pairs from the experiment, which covered the first three vibration modes of the ring. These modes are located at approximately **70.8 Hz (Mode 1)**, **200.4 Hz (Mode 2)**, and **386.7 Hz (Mode 3)**.

However, as noted in the experimental difficulties, 11 of these 18 pairs were invalidated because they did not meet the fundamental physical condition $f_S < f_O$. This issue affected *all* measurements for Mode 2, and several measurements for Modes 1 and 3 from the “Open All” test set.

Consequently, these 11 invalid pairs are omitted from the results table, as no k_c value could be calculated for them. The analysis of the 7 remaining, validated data pairs is presented in Table 2, which consistently showed a low electromechanical coupling factor.

Table 2: Equivalent Electrical Parameters for Valid Test Configurations

Pair ID	Test Configuration	Mode	f_O (Hz)	f_S (Hz)	k_c (%)	L (H)	R (Ω)
1	Open A / Short All	1	70.85	70.83	2.20	37.38	448.8
3	Open A / Short All	3	386.73	386.57	2.80	1.25	104.7
4	Open B / Short All	1	70.84	70.83	1.07	49.00	287.6
6	Open B / Short All	3	386.76	386.57	3.12	1.64	152.62
7	Open C / Short All	1	70.84	70.83	0.88	37.96	182.70
9	Open C / Short All	3	386.79	386.57	3.31	1.273	125.5
10	Open All / Short A	1	70.86	70.85	1.14	37.37	232.2

Note:

The 7 valid pairs are listed above. The calculated k_c values are consistently below 5%, validating the observed limitation.

6 Understanding

The parameters k_c , L , and R provide a complete electrical analogy for the mechanical mode of vibration.

- **Electromechanical Coupling Factor (k_c):** This dimensionless factor quantifies the efficiency with which energy is converted between the mechanical and electrical domains. A higher k_c means that more mechanical energy can be captured and transferred to an electrical shunt circuit. The consistently low values (0.88% – 3.31%) are a critical finding. This low coupling is likely the result of non-optimal patch placement (e.g., placing the patch near a node of vibration), or, more plausibly, the compounding effect of the faulty patch connection, which effectively reduced the active area and, therefore, the overall coupling of the system.
- **Equivalent Inductance (L):** The inductance L in the electrical model is the analog of the effective modal mass of the mechanical system. A larger L corresponds to a lower resonant frequency (f_O) when C^e is constant. This relationship ($L \propto 1/C^e$) is why the L values vary in the table. Since the f_O for Mode 1 is almost constant (≈ 70.85 Hz), the **lower capacitance of Batch B (103 nF) necessarily results in a larger calculated inductance (49.00 H)** compared to Batch A (37.38 H) or C (37.96 H). This pattern holds for Mode 3 as well.
- **Equivalent Resistance (R):** The resistance R is the analog of the mechanical damping (viscosity) of the structure. The lower R for the higher mode (Mode 3, ≈ 105 – 153Ω) compared to Mode 1 (≈ 183 – 449Ω) suggests that the damping mechanism (whether intrinsic material damping or external friction) has a frequency-dependent nature, although the variability of the results suggests caution in interpretation.

Despite the low coupling factor, the successful identification of the L and R parameters for the valid modes confirms that the R-L-C circuit model is a viable representation of the ring’s dynamic behavior, providing the necessary data for designing a R-L shunt circuit that could theoretically damp the target modes.

7 Further Analysis and Conclusion

The parameters calculated in this experiment serve as the essential first step for designing a vibration damping system. However, the results and the experimental challenges highlight several key practical considerations that align with state-of-the-art research in this field.

7.1 The Challenge of Low-Frequency Inductance

A primary “out-of-the-box” finding is the magnitude of the calculated equivalent inductance (L). The values for Mode 1 (e.g., 37.38 H and 49.00 H) are extremely large for standard electronic components. This is not an error, but a well-known challenge in passive shunt damping.

Research on this topic, including a study on a similar piezoelectric ring by Luo et al. [1], confirms that “the inductance requirements may be extremely high when targeting low frequencies, in the thousands of Henrys.” This practical constraint often makes passive shunts with physical inductors unfeasible for low-frequency modes. The engineering solution, noted in the literature, is to use “synthetic inductors” active operational-amplifier circuits that electronically simulate the required high inductance. The calculated R value, representing the optimal resistance for energy dissipation, would then be implemented using a standard resistor in series with this synthetic inductor to complete the tuned damping circuit.

7.2 Adapting to Real-World Imperfections

This experiment was significantly impacted by a “faulty connection” in Batch B, an example of real-world hardware imperfection. This challenge is not unique to a teaching lab; the authors of the Luo et al. study also faced significant deviations from their ideal model, including an “imperfect mechanical ring” that was elliptical and non-ideal transformer components.

Their solution was not to discard the experiment, but to apply “numerical optimization” to adapt their electrical network to the “imperfect mechanical structure.” This demonstrates a key engineering principle: the design process must be robust enough to adapt to and correct for real-world imperfections, which our own capacitance measurements for Batch B (103 nF) clearly identified.

7.3 Conclusion and System Limitations

This experiment successfully characterized the first and third vibration modes of the 15-patch piezoelectric ring, yielding the necessary parameters (k_c, L, R) for a single-mode resonant shunt.

However, the 15-patch setup itself has implications for more advanced damping. The Luo et al. study, which used a 15-cell network to damp multiple modes, concluded that this number was “insufficient to achieve high attenuation for all three modes” and that a minimum of 40 cells might be needed. This suggests that while our 15-patch ring is suitable for this characterization and for simple single-mode damping, a more effective broadband (multi-modal) damping system would require a significantly higher number of patches.

References

- [1] Luo, A., Lossouarn, B., & Erturk, A. (2023). Analogous piezoelectric network for multimodal vibration attenuation of a thin circular ring. *Smart Materials and Structures*, 32(11), 115024.